E1 - Healthz (servicio vivo)

```
jazmin@ubuntu-healthchecks:~$ curl -s http://localhost:8001/healthz | python3 -m json.tool
{
    "status": "alive"
}
```

Comando: curl -s http://localhost:8001/healthz
Resultado: {"status": "alive"}
Requisito TP2 cumplido: B2 - "GET /healthz (vivo)"

E2 - Readyz (BD conectada)

```
jazmin@ubuntu-healthchecks:~$ curl -s http://localhost:8001/readyz | python3 -m json.tool
{
    "status": "ready"
}
```

Comando: curl -s http://localhost:8001/readyz
Resultado: {"status": "ready"}
Requisito TP2 cumplido: B2 - "GET /readyz (listo: sus dependencias responden), separados"

E3 - Métricas Prometheus

```
jazmin@ubuntu-healthchecks:~$ curl -s http://localhost:8001/metrics | head -20
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 0.0
python_gc_objects_collected_total{generation="1"} 355.0
python_gc_objects_collected_total{generation="2"} 0.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 0.0
python_gc_collections_total{generation="1"} 12.0
python_gc_collections_total{generation="2"} 0.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="14",patchlevel="4",version="3.14.4"}
1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
```

Comando: curl -s http://localhost:8001/metrics | head -20
Requisito TP2 cumplido: B2 - "GET /metrics en formato Prometheus"
Expone http_requests_total y http_request_duration_seconds

E4 - Contrato idéntico al monolito (Strangler Fig)
(curl al puerto 8000 - monolito)

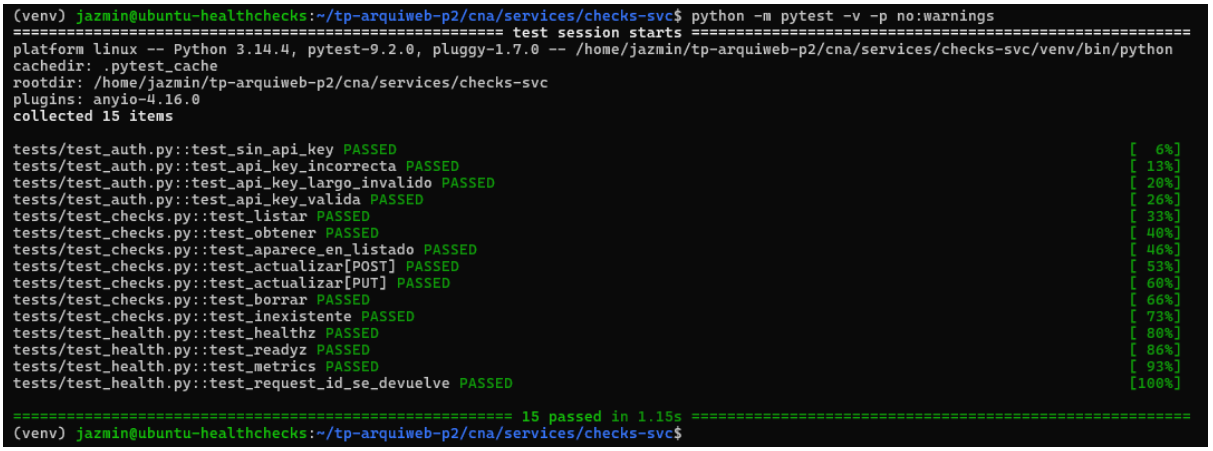
```
Last login: Thu Nov 12 23:57:38 2026 from 10.0.2.2
jazmin@ubuntu-healthchecks:~$ curl -s -H "X-Api-Key: 740d2902981530e37bc76e978cf20156" http://localhost:8000/api/v3/checks/ |
 python3 -m json.tool
{
    "checks": [
        {
            "name": "My First Check",
            "slug": "my-first-check",
            "tags": "",
            "desc": "",
            "grace": 3600,
            "n_pings": 0,
            "status": "new",
            "started": false,
            "last_ping": null,
            "next_ping": null,
            "manual_resume": false,
            "methods": "",
            "subject": "",
            "subject_fail": "",
            "start_kw": "",
            "success_kw": "",
            "failure_kw": "",
            "filter_subject": false,
            "filter_body": false,
            "filter_http_body": false,
            "filter_default_fail": false,
            "badge_url": "http://localhost:8000/b/2/b81e24b3-d7a3-45d3-a03b-0e6a1ad727c3.svg",
            "uuid": "306e103d-7fe3-4519-85a6-942169a19aa7",
            "ping_url": "http://localhost:8000/ping/306e103d-7fe3-4519-85a6-942169a19aa7",
            "update_url": "http://localhost:8000/api/v3/checks/306e103d-7fe3-4519-85a6-942169a19aa7",
            "pause_url": "http://localhost:8000/api/v3/checks/306e103d-7fe3-4519-85a6-942169a19aa7/pause",
            "resume_url": "http://localhost:8000/api/v3/checks/306e103d-7fe3-4519-85a6-942169a19aa7/resume",
            "channels": "1857d9cc-3af6-4636-9d8c-8274d370068e",
            "timeout": 86400
        }
    ]
}
```

(curl al puerto 8001 - microservicio)

```
jazmin@ubuntu-healthchecks:~$ curl -s -H "X-Api-Key: 740d2902981530e37bc76e978cf20156" http://localhost:8001/api/v3/checks/ | python3 -m json.tool
{
    "checks": [
        {
            "name": "My First Check",
            "slug": "my-first-check",
            "tags": "",
            "desc": "",
            "grace": 3600,
            "n_pings": 0,
            "status": "new",
            "started": false,
            "last_ping": null,
            "next_ping": null,
            "manual_resume": false,
            "methods": "",
            "subject": "",
            "subject_fail": "",
            "start_kw": "",
            "success_kw": "",
            "failure_kw": "",
            "filter_subject": false,
            "filter_body": false,
            "filter_http_body": false,
            "filter_default_fail": false,
            "badge_url": "http://localhost:8001/b/2/b81e24b3-d7a3-45d3-a03b-0e6a1ad727c3.svg",
            "uuid": "306e103d-7fe3-4519-85a6-942169a19aa7",
            "ping_url": "http://localhost:8001/ping/306e103d-7fe3-4519-85a6-942169a19aa7",
            "update_url": "http://localhost:8001/api/v3/checks/306e103d-7fe3-4519-85a6-942169a19aa7",
            "pause_url": "http://localhost:8001/api/v3/checks/306e103d-7fe3-4519-85a6-942169a19aa7/pause",
            "resume_url": "http://localhost:8001/api/v3/checks/306e103d-7fe3-4519-85a6-942169a19aa7/resume",
            "channels": "",
            "timeout": 86400
        }
    ]
}
```

El microservicio devuelve el mismo JSON que el monolito para el mismo check (UUID 306e103d-...).
Esto prueba que puede reemplazar al monolito sin romper clientes.

E5 - Pruebas automatizadas (15 pruebas con un solo comando)



Comando: python -m pytest -v -p no:warnings
Resultado: 15 passed (autenticación 4, CRUD de checks 7, salud y métricas 4)
Requisito TP2 cumplido: B2 - "al menos 3 pruebas automatizadas ejecutables con un solo comando"

E6 - Crear un check (POST)



Comando: curl -s -X POST -H "X-Api-Key: ..." -H "Content-Type: application/json" -d '{"name":"Evidencia TP2",...}' http://localhost:8001/api/v3/checks/
Resultado: el servicio devuelve el check creado, con su uuid, nombre "Evidencia TP2" y timeout 3600
Requisito TP2 cumplido: B2 - replicar el endpoint de creación del monolito (Strangler Fig)

E7 - Obtener un check por UUID (GET)

```
(venv) jazmin@ubuntu-healthchecks:~/tp-arquiweb-p2/cna/services/checks-svc$ curl -s -H "X-Api-Key: 740d2902
981530e37bc76e978cf20156" http://localhost:8001/api/v3/checks/$UUID | python3 -m json.tool
{
    "name": "Evidencia TP2",
    "slug": "evidencia-tp2",
    "tags": "test",
    "desc": "",
    "grace": 600,
    "n_pings": 0,
    "status": "new",
    "started": false,
    "last_ping": null,
    "next_ping": null,
    "manual_resume": false,
    "methods": "",
    "subject": "",
    "subject_fail": "",
    "start_kw": "",
    "success_kw": "",
    "failure_kw": "",
    "filter_subject": false,
    "filter_body": false,
    "filter_http_body": false,
    "filter_default_fail": false,
    "badge_url": "http://localhost:8001/b/2/49b4de20-d05d-4908-9254-68deba34484c.svg",
    "uuid": "62460832-6537-461a-94af-6d6b4f5a435d",
    "ping_url": "http://localhost:8001/ping/62460832-6537-461a-94af-6d6b4f5a435d",
    "update_url": "http://localhost:8001/api/v3/checks/62460832-6537-461a-94af-6d6b4f5a435d",
    "pause_url": "http://localhost:8001/api/v3/checks/62460832-6537-461a-94af-6d6b4f5a435d/pause",
    "resume_url": "http://localhost:8001/api/v3/checks/62460832-6537-461a-94af-6d6b4f5a435d/resume",
    "channels": "",
    "timeout": 3600
}
(venv) jazmin@ubuntu-healthchecks:~/tp-arquiweb-p2/cna/services/checks-svc$
```

Comando: curl -s -H "X-Api-Key: ..." http://localhost:8001/api/v3/checks/<uuid>
Resultado: devuelve el mismo check creado en E6
Requisito TP2 cumplido: B2 - replicar el endpoint de lectura del monolito

E8 - Actualizar un check (POST)

```
(venv) jazmin@ubuntu-healthchecks:~/tp-arquiweb-p2/cna/services/checks-svc$ curl -s -X POST -H "X-Api-Key:
740d2902981530e37bc76e978cf20156" -H "Content-Type: application/json" -d '{"name":"Evidencia Modificada","t
imeout":7200}' http://localhost:8001/api/v3/checks/$UUID | python3 -m json.tool
{
    "name": "Evidencia Modificada",
    "slug": "evidencia-tp2",
    "tags": "test",
    "desc": "",
    "grace": 600,
    "n_pings": 0,
    "status": "new",
    "started": false,
    "last_ping": null,
    "next_ping": null,
    "manual_resume": false,
    "methods": "",
    "subject": "",
    "subject_fail": "",
    "start_kw": "",
    "success_kw": "",
    "failure_kw": "",
    "filter_subject": false,
    "filter_body": false,
    "filter_http_body": false,
    "filter_default_fail": false,
    "badge_url": "http://localhost:8001/b/2/49b4de20-d05d-4908-9254-68deba34484c.svg",
    "uuid": "62460832-6537-461a-94af-6d6b4f5a435d",
    "ping_url": "http://localhost:8001/ping/62460832-6537-461a-94af-6d6b4f5a435d",
    "update_url": "http://localhost:8001/api/v3/checks/62460832-6537-461a-94af-6d6b4f5a435d",
    "pause_url": "http://localhost:8001/api/v3/checks/62460832-6537-461a-94af-6d6b4f5a435d/pause",
    "resume_url": "http://localhost:8001/api/v3/checks/62460832-6537-461a-94af-6d6b4f5a435d/resume",
    "channels": "",
    "timeout": 7200
}
(venv) jazmin@ubuntu-healthchecks:~/tp-arquiweb-p2/cna/services/checks-svc$
```

Comando: curl -s -X POST ... -d '{"name":"Evidencia Modificada","timeout":7200}' http://localhost:8001/api/v3/checks/<uuid>
Resultado: el nombre pasó a "Evidencia Modificada" y el timeout a 7200
Requisito TP2 cumplido: B2 - replicar el endpoint de actualización del monolito

E9 - Borrar un check (DELETE)

```
(venv) jazmin@ubuntu-healthchecks:~/tp-arquiweb-p2/cna/services/checks-svc$ curl -s -X DELETE -H "X-Api-Key
: 740d2902981530e37bc76e978cf20156" http://localhost:8001/api/v3/checks/$UUID | python3 -m json.tool
{
    "ok": true
}
(venv) jazmin@ubuntu-healthchecks:~/tp-arquiweb-p2/cna/services/checks-svc$
```

Comando: curl -s -X DELETE -H "X-Api-Key: ..." http://localhost:8001/api/v3/checks/<uuid>
Resultado: {"ok": true}
Requisito TP2 cumplido: B2 - replicar el endpoint de borrado del monolito
Nota: la respuesta del monolito devuelve el check borrado; checks-svc devuelve {"ok": true}. El código de estado y el efecto son los mismos.

E10 - Listar checks tras el borrado (GET)

```
(venv) jazmin@ubuntu-healthchecks:~/tp-arquiweb-p2/cna/services/checks-svc$ curl -s -H "X-Api-Key: 740d2902
981530e37bc76e978cf20156" http://localhost:8001/api/v3/checks/ | python3 -m json.tool
{
    "checks": [
        {
            "name": "My First Check",
            "slug": "my-first-check",
            "tags": "",
            "desc": "",
            "grace": 3600,
            "n_pings": 0,
            "status": "new",
            "started": false,
            "last_ping": null,
            "next_ping": null,
            "manual_resume": false,
            "methods": "",
            "subject": "",
            "subject_fail": "",
            "start_kw": "",
            "success_kw": "",
            "failure_kw": "",
            "filter_subject": false,
            "filter_body": false,
            "filter_http_body": false,
            "filter_default_fail": false,
            "badge_url": "http://localhost:8001/b/2/b81e24b3-d7a3-45d3-a03b-0e6a1ad727c3.svg",
            "uuid": "306e103d-7fe3-4519-85a6-942169a19aa7",
            "ping_url": "http://localhost:8001/ping/306e103d-7fe3-4519-85a6-942169a19aa7",
            "update_url": "http://localhost:8001/api/v3/checks/306e103d-7fe3-4519-85a6-942169a19aa7",
            "pause_url": "http://localhost:8001/api/v3/checks/306e103d-7fe3-4519-85a6-942169a19aa7/pause",
            "resume_url": "http://localhost:8001/api/v3/checks/306e103d-7fe3-4519-85a6-942169a19aa7/resume"
,
            "channels": "",
            "timeout": 86400
        }
    ]
}
(venv) jazmin@ubuntu-healthchecks:~/tp-arquiweb-p2/cna/services/checks-svc$
```

Comando: curl -s -H "X-Api-Key: ..." http://localhost:8001/api/v3/checks/
Resultado: solo aparece "My First Check"; "Evidencia Modificada" ya no está
Requisito TP2 cumplido: B2 - el borrado se refleja en la base de datos compartida

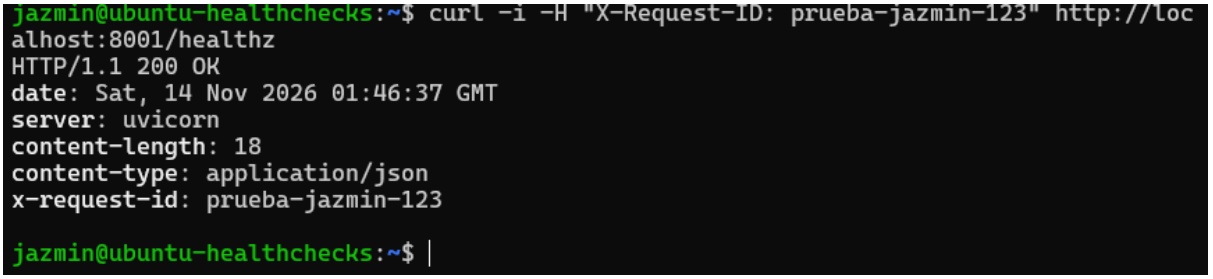
E11 - Logs en formato JSON a stdout

```
(venv) jazmin@ubuntu-healthchecks:~/tp-arquiweb-p2/cna/services/checks-svc$ uvicorn app.main:app --host 0.0
.0.0 --port 8001
INFO:     Started server process [2652]
INFO:     Waiting for application startup.
{"timestamp": "2026-11-14T01:44:10.080855+00:00", "level": "INFO", "service": "checks-svc", "request_id": "
-", "message": "checks-svc started"}
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8001 (Press CTRL+C to quit)
INFO:     127.0.0.1:55112 - "GET /healthz HTTP/1.1" 200 OK
```

Comando: uvicorn app.main:app --host 0.0.0.0 --port 8001

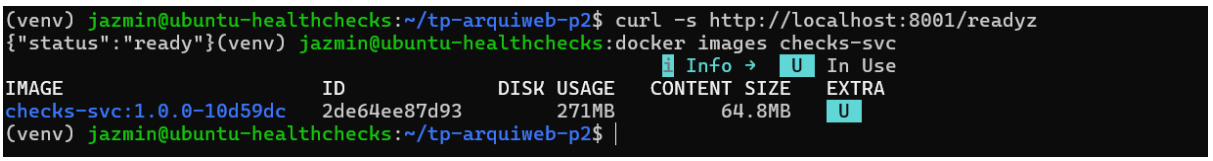
Resultado: el servicio escribe sus logs como JSON con timestamp, level, service, request_id y message (ejemplo: "checks-svc started")
Requisito TP2 cumplido: B2 - "logs estructurados en JSON a stdout"

E12 - Propagación de X-Request-ID



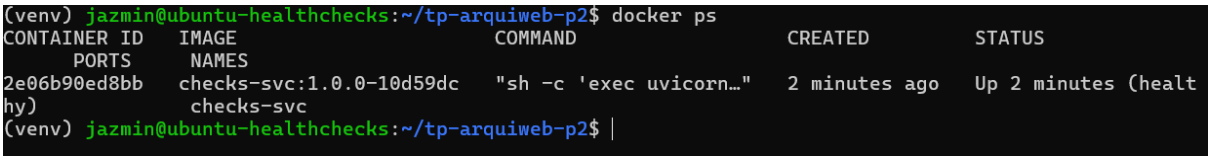
Comando: curl -i -H "X-Request-ID: prueba-jazmin-123" http://localhost:8001/healthz
Resultado: la respuesta devuelve el encabezado x-request-id con el mismo valor enviado (prueba-jazmin-123)
Requisito TP2 cumplido: B2 - "propagar X-Request-ID"

E13 - Imagen Docker de checks-svc (B3)



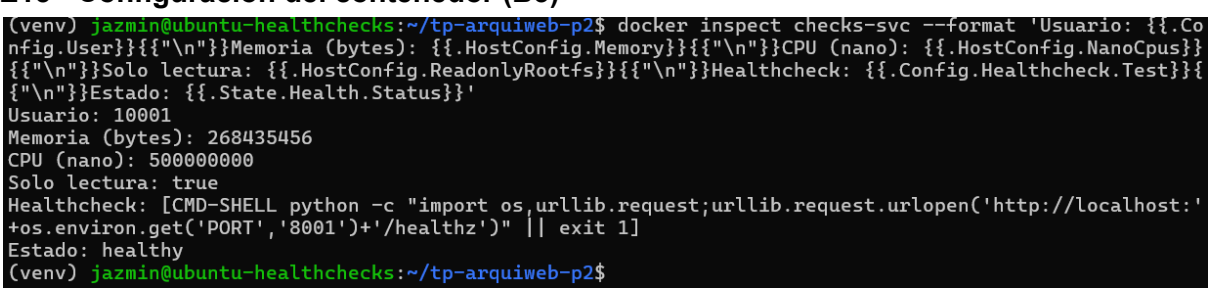
Comando: docker images checks-svc
Resultado: imagen checks-svc con etiqueta 1.0.0-10d59dc (versión + hash corto del commit), 64,8 MB
Requisito TP2 cumplido: B3 - "imagen etiquetada con versión + hash corto del commit; prohibido :latest"

E14 - Contenedor en ejecución (B3)



Comando: docker ps
Resultado: contenedor checks-svc "Up" y "(healthy)": el chequeo de salud responde
Requisito TP2 cumplido: B3 - HEALTHCHECK funcionando.

E15 - Configuración del contenedor (B3)



Comando: docker inspect checks-svc --format '...'

Resultado: usuario 10001 (no root), memoria limitada a 256 MB, CPU limitada a 0,5, sistema de archivos de solo lectura, healthcheck declarado y en estado healthy
Requisito TP2 cumplido: B3 - usuario no root, límites de CPU y memoria, read_only y HEALTHCHECK